

11. (2.5 points)

Given the following information:

- Cumulative reported losses at 24 months for accident year 2015 are \$9,000
- The following reported development factors were derived:

|            | Loss<br>Development | Cumulative<br>Development |                   |
|------------|---------------------|---------------------------|-------------------|
| <u>Age</u> | <u>Factor</u>       | <u>Factor</u>             | <u>% Reported</u> |
| 12         | 2.000               | 3.000                     | 33.3%             |
| 24         | 1.364               | 1.500                     | 66.7%             |
| 36         | 1.073               | 1.100                     | 90.9%             |
| 48         | 1.025               | 1.025                     | 97.6%             |
| 60         | 1.000               | 1.000                     | 100.0%            |

- Incremental losses,  $C_{ij}$ , follow an over-dispersed Poisson distribution with mean  $x_i y_i$  and variance  $\phi x_i y_i$
- The variable  $x_i$  represents the expected ultimate losses for accident year  $i$ .
- The variable  $y_j$  represents the proportion of ultimate losses that emerge in development year  $j$ .
- The prior distribution for  $x_i$  is gamma with mean  $\alpha_i / \beta_i$  and variance  $\alpha_i / \beta_i^2$
- The dispersion parameter,  $\phi$ , for the over-dispersed Poisson distribution is 9.125
- The accident year 2015 estimates for  $\alpha$  and  $\beta$  are 100 and 0.01234 respectively.
- The mean of  $C_{ij}$  for this Bayesian model is:

$$Z_{ij} (\lambda_j - 1) D_{i,j-1} + (1 - Z_{ij}) (\lambda_j - 1) M_i \frac{1}{\lambda_j \lambda_{j+1} \dots \lambda_n}, \text{ where } Z_{ij} = \frac{\sum_{k=1}^{j-1} y_k}{\beta_i \phi + \sum_{k=1}^{j-1} y_k}$$

- $\lambda_j$  is the incremental chain ladder loss development factor for development year  $j$ .
- $D_{ij}$  is the cumulative losses for accident year  $i$  as of development year  $j$ .

a. (2 points)

Calculate the incremental losses for accident year 2015 expected to emerge between 24 and 48 months of development using the model.

b. (0.5 point)

Identify and briefly describe what parameter in the model would have to change in order to produce IBNR estimates closer to chain ladder indications.

**CONTINUED ON NEXT PAGE**

**EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT**

| <b>QUESTION 11</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |
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| <b>TOTAL POINT VALUE: 2.5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>LEARNING OBJECTIVE(S): A10</b> |
| <b>SAMPLE ANSWERS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |
| <b>Part a: 2 points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |
| <p><u>Sample 1</u><br/> 24 to 36 months:<br/> <math>Z_{ij} = (.667) / (0.01234 * 9.125 + .667) = .856</math>    <math>\lambda_j = 1.364</math>    <math>D_{ij-1} = 9000</math>    <math>M = \alpha / \beta = 8104</math><br/> % emerged = <math>.909 - .667 = .242</math><br/> <math>E[C_{ij}] = .856 * 9000 * .364 + (1-.856) * 8104 * .242 = 3087</math></p> <p>36 to 48 months:<br/> <math>Z_{ij} = (.909) / (0.01234 * 9.125 + .909) = .890</math>    <math>\lambda_j = 1.073</math>    <math>D_{ij-1} = 12087</math><br/> % emerged = <math>.067</math><br/> <math>E[C_{ij}] = .89 * 12087 * .073 + (1-.89) * 8104 * .067 = 845</math></p> <p><math>3087 + 845 = 3932</math></p> <p><u>Sample 2</u><br/> <math>Z = (1/CDF) / (\phi\beta + 1/CDF) = 1/1.5 / (9.125(0.1234) + 1/1.5) = .856</math><br/> <math>E(C_{2015\ 36}) = .856(1.364 - 1)(9000) + (1-.856)(1.364-1)(8130.7)(1/1.5) = 3087.4</math><br/> <math>M = \alpha / \beta = 100 / .01234 = 8103.7</math></p> <p><math>C_{2015\ 48} = 1/1.1 / (9.125(0.1234) + 1/1.1) = .890</math><br/> <math>E(C_{2015\ 48}) = .890(1.073-1)(9000+3087.4) + (1-.890)(1.073-1)(8103.7)(1/1.1) = 844.5</math></p> <p>Total incr. emergence 24-48 mo = <math>3087.4 + 844.5 = 3931.9</math></p> <p><u>Sample 3</u><br/> <math>E[C_{15,\ 24-48}] = (.667 / (.667 + .01234(9.125))) (1.364*1.073 - 1) (9000) + (1 - .8556) ((1.364*1.073 - 1) / 1.5) (8103.7277) = 3931.3311</math></p> <p><u>Sample 4</u><br/> <math>E(\text{Incr Loss CL}) = 9000(1.5)(.976-.667) = 4171.5</math><br/> <math>E(\text{Incr Loss prior}) = 100/.01234 (.976 - .667) = 2504</math></p> <p><math>Z_{ij} = \% \text{ reported} / \phi\beta + \% \text{ reported} = .667 / (9.125(.01234) + .667) = .856</math></p> <p>Expected incremental = <math>.856(4171.5) + (1-.856)(2504) = 3931.4</math></p> |                                   |
| <b>Part b: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |
| <p><u>Sample 1</u><br/> <math>\beta</math> would have to change. <math>\beta</math> controls the variance for the prior distribution. If we want an estimate closer to chain ladder, we want a larger variance in the prior distribution so we want a smaller <math>\beta</math>.</p> <p><u>Sample 2</u><br/> If <math>\beta</math> is reduced, it means Z gets larger and more weight is put on the CL indications in the Bayesian</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |

## EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

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| model formula.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>EXAMINER'S REPORT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Candidates were expected to demonstrate an understanding of both the incremental chain ladder estimates and incremental B-F estimates for IBNR and use credibility weights to estimate 2 years IBNR development. Candidates were also expected to understand the key parameters used in the model.                                                                                                                                                                                                                                                                                                  |
| <b>Part a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>Candidates were expected to obtain estimates for IBNR in 2017 and 2018</p> <p>Common errors include:</p> <ul style="list-style-type: none"><li>• Omitting the 36 to 48 month development</li><li>• Using chain ladder estimate for <math>C_{2015,36}</math> in place of credibility weighted IBNR from the model</li><li>• Applying incorrect Loss Development factors (Chain Ladder or B-F)</li><li>• Applying an incorrect percent reported (in the credibility calculation and in the B-F estimate)</li></ul>                                                                                 |
| <b>Part b</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>Candidates were expected to show understanding of the model parameters which changed the weights between the chain ladder method and the Bornhuetter-Ferguson method of estimating IBNR.</p> <p>Common errors include:</p> <ul style="list-style-type: none"><li>• Indicating the wrong direction for <math>\beta</math></li><li>• Providing direction only for dispersion parameter (<math>\phi</math>) - text provides examples of actuary selecting <math>\beta</math> and refers to <math>\phi</math> as a "nuisance parameter" that is solved using maximum likelihood estimates.</li></ul> |